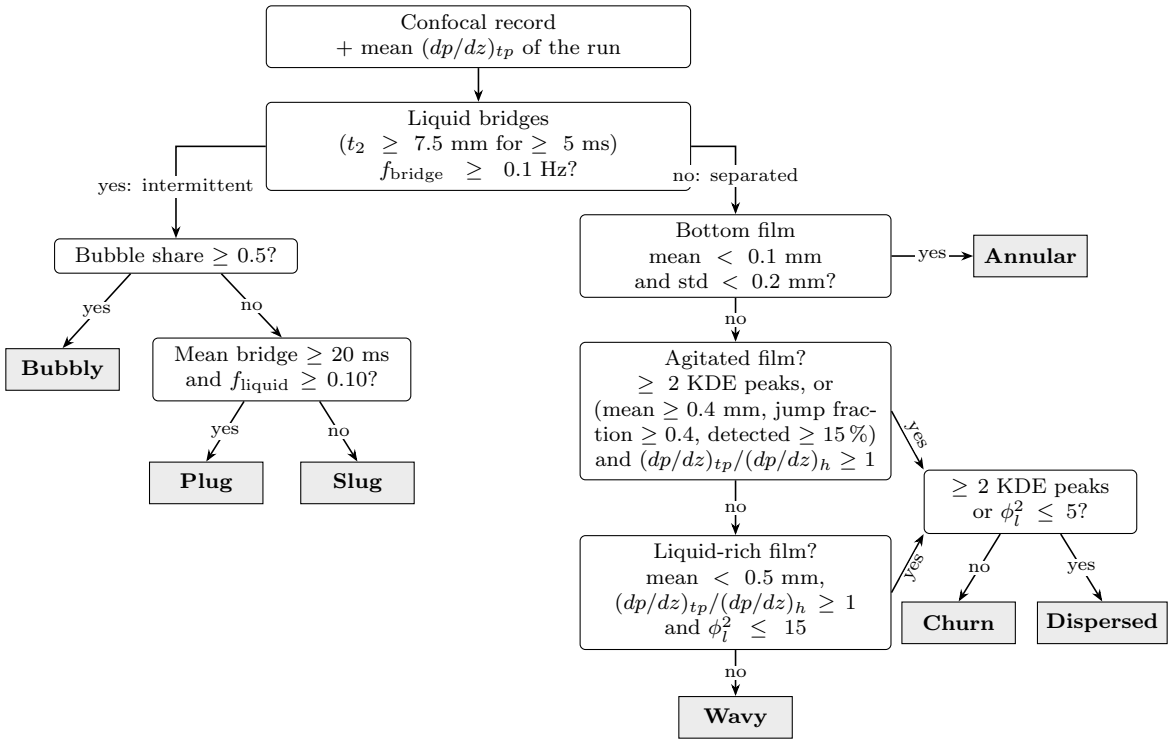

Confocal record
+ mean $(dp/dz)_{tp}$ of the run
Liquid bridges
($t_2 \geq 7.5$ mm for ≥ 5 ms)
$f_{\text{bridge}} \geq 0.1$ Hz?
yes: intermittent
no: separated
Bubble share ≥ 0.5?
yes
no
Bubbly
Mean bridge ≥ 20 ms
and $f_{\text{liquid}} \geq 0.10$?
yes
no
Plug
Slug
Bottom film
mean $<$ 0.1 mm
and std $<$ 0.2 mm?
yes
Annular
no
Agitated film?
$\geq$ 2 KDE peaks, or
(mean ≥ 0.4 mm, jump fraction ≥ 0.4, detected $\geq 15\,\%$)
and $(dp/dz)_{tp}/(dp/dz)_h \geq 1$
yes
$\geq$ 2 KDE peaks
or $\phi_l^2 \leq 5$?
no
Liquid-rich film?
mean $<$ 0.5 mm,
$(dp/dz)_{tp}/(dp/dz)_h \geq 1$
and $\phi_l^2 \leq 15$
yes
no
Churn
yes
Dispersed
no
Wavy